

$$Q.1) |E(uv)| \leq \sqrt{E(u^2) E(v^2)}$$

$\alpha u = \beta v = 1$ for some real α and β that are not both equal to zero.

Let us consider the random variable $Y = (u - \alpha v)^2$, Y is a positive random variable for all $\alpha \in \mathbb{R}$.

$$0 \leq E(Y)$$

$$0 \leq E(u - \alpha v)^2$$

$$0 \leq E(u^2 - 2\alpha uv + \alpha^2 v^2)$$

$$0 \leq E(u^2) - 2\alpha E(uv) + \alpha^2 E(v^2)$$

taking the discriminant:

$$(-2E(uv))^2 - 4E(v^2)E(u^2) \leq 0$$

$$E(uv)^2 - E(v^2)E(u^2) \leq 0$$

$$E(uv)^2 \leq E(v^2)E(u^2)$$

taking underoot on both sides:

$$|E(uv)| \leq \sqrt{E(u^2) E(v^2)}$$

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$$\textcircled{2} E[(u - E(u))(v - E(v))] = \text{cov}(u, v)$$

||

$$= E[uv - uE(v) - vE(u) + E(u)E(v)]$$

$$= E[uv] - E[uE(v)] - E[vE(u)] + E[E(u)E(v)]$$

$$= E[u]E[v] - E[u]E[v] - E[v]E[u] + E[u]E[v]$$

$$= E[uv] - E[u]E[v]$$

$$= \text{cov}(u, v)$$